

Simulation studies of the ePIC Silicon Vertex Tracker

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Motivation

- Tracking: Reconstructing path of particles from detector hits
- Allows for measurement of momentum and vertex positions
- Detector design decisions affect resolution so early simulations are very important
- The ePIC (Electron-Proton/Ion Collider) experiment needs a high precision silicon tracking detector

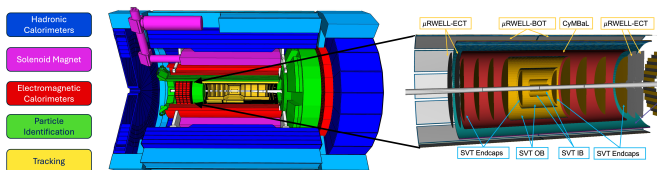
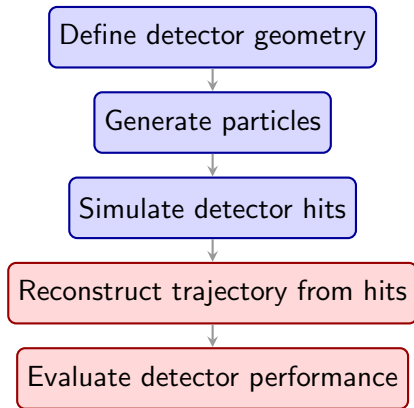


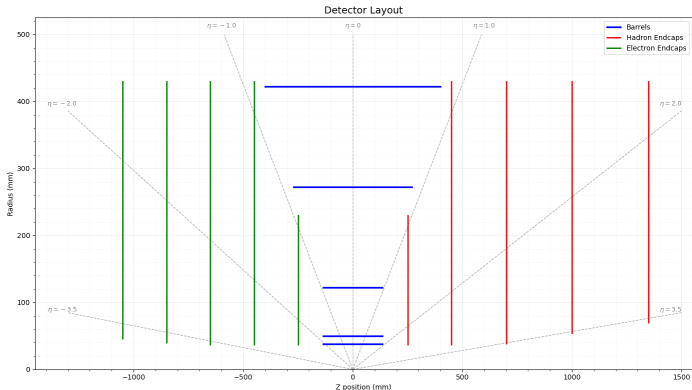
Figure: ePIC central detector, with tracking detectors highlighted

Project overview

- Aim: Construct a simplified simulation of the ePIC Silicon Vertex Tracker (SVT), and evaluate the tracking performance for different detector parameters
- Simulations use Geant4, a C++ toolkit for simulating the passage of particles through matter using Monte Carlo methods



Detector geometry



- 3 Inner Barrel layers, 2 Outer Barrel layers, 10 endcap discs
- Also include copper layers to simulate support material, beryllium beampipe, and 1.7 T uniform magnetic field

Particle generation

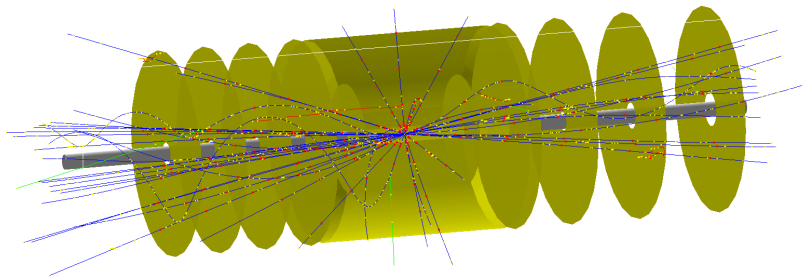


Figure: Screenshot of Geant4 simulation of detector with 50 pions

- Positive pions were generated one at a time from the origin
- Pions chosen as they are the most common final state charged particle for electron-proton collisions at these energies

Detector hits

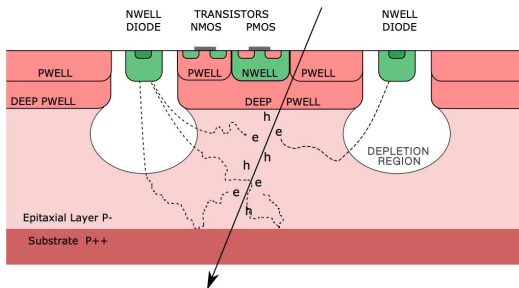
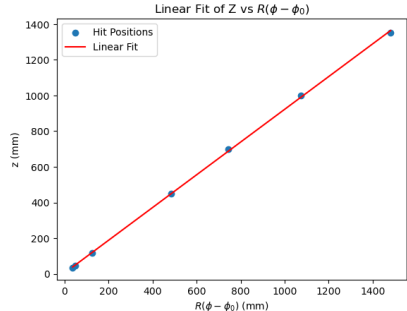
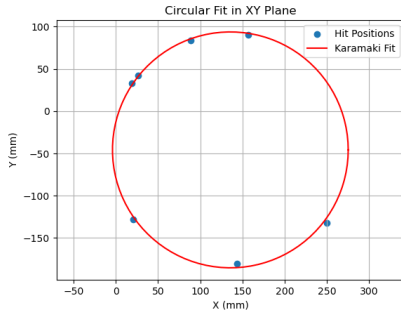


Figure: Schematic diagram of a Monolithic Active Pixel Sensor

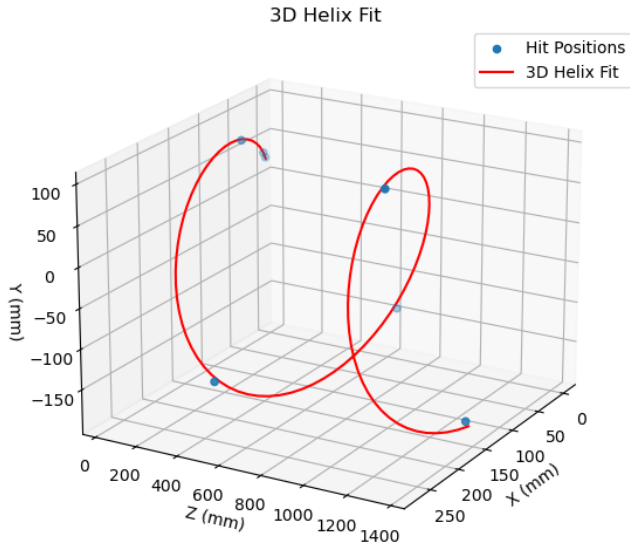
- Only register hits from primary pions, no secondary particles
- Apply a random Gaussian smearing to the hit position, with uncertainty $\sigma = 7 \mu\text{m}$, to simulate effect of finite pixel size, charge diffusion and pixel clustering

Track reconstruction

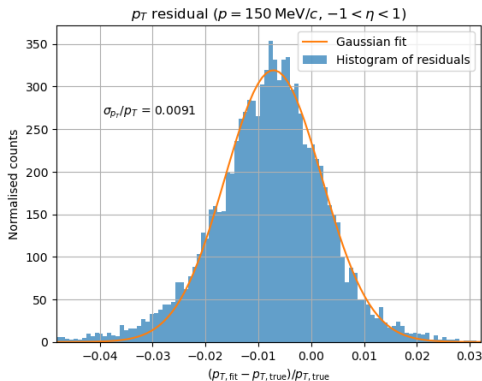


- Charged particles in a magnetic field move in a helical trajectory
- Fit a circle to the xy -positions, and linearly fit z against the azimuthal angle of the momentum
- From these fitted parameters, we can determine the original position and momentum

Track reconstruction



Measuring tracking resolution

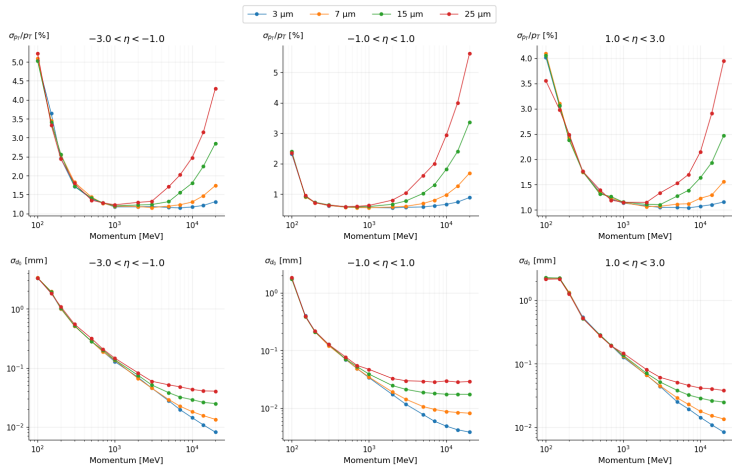


Residuals:

- Momentum: $\frac{p_{\text{fit}} - p_{\text{true}}}{p_{\text{true}}}$
- Position: $x_{\text{fit}} - x_{\text{true}}$

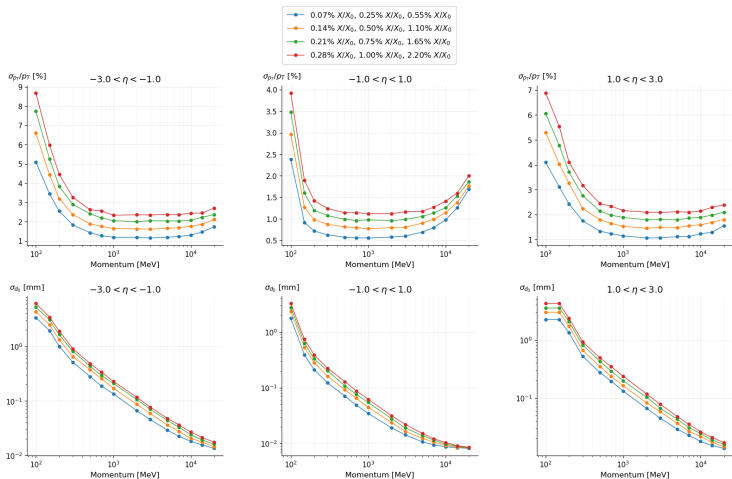
Simulate $n=5,000,000$ pion tracks, plot histogram of residual, fit Gaussian and extract uncertainty

Tracking resolution vs hit resolution



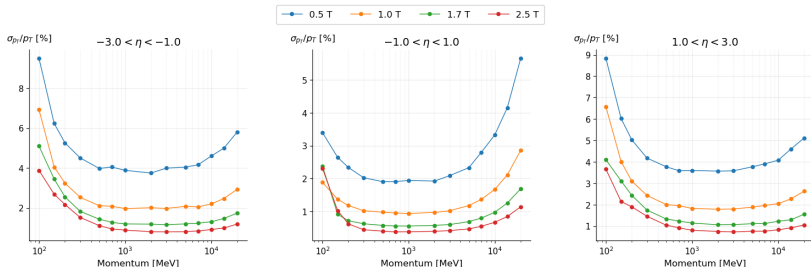
Intrinsic detector uncertainty affects resolution at large momentum

Tracking resolution vs material thickness



Material thickness strongly affects resolution across all momenta

Tracking resolution vs magnetic field



Magnetic field affects momentum resolution, but not position resolution

- More realistic geometry, magnetic field and hit digitisation model
- Use track finding algorithms to study realistic collision scenarios (matching hits to tracks)
- Test more sophisticated tracking algorithms e.g. Kalman Filters
- Investigate alterations to detector geometry, such as changing number, position, or shape of layers